

Multi-Shop Commerce Software – Database Structure (Step by Step)

এই ডাটাবেস স্ট্রাকচারটি **দোকান + অনলাইন + মাল্টি-শপ** কমার্স সফটওয়্যারের জন্য। ধাপে ধাপে সাজানো হয়েছে যেন আপনি সহজে বুঝে ইমপ্লিমেন্ট করতে পারেন।

STEP 1: Super Admin & Shop (Multi-Tenant Base)

Shops

প্রতিটি দোকানের আলাদা পরিচয়

Field	Type	Description
ShopId	INT (PK)	দোকানের ইউনিক আইডি
ShopName	NVARCHAR(200)	দোকানের নাম
OwnerName	NVARCHAR(150)	মালিকের নাম
Mobile	NVARCHAR(20)	মোবাইল
Email	NVARCHAR(100)	ইমেইল
IsActive	BIT	চালু / বন্ধ
CreatedDate	DATETIME	তৈরির তারিখ

 সব টেবিলে ShopId থাকবে (Multi-Shop rule)

STEP 2: User & Role Management

Users

Field	Type
UserId	INT (PK)
ShopId	INT (FK)
UserName	NVARCHAR(100)
PasswordHash	NVARCHAR(MAX)

Field	Type
RoleId	INT
IsActive	BIT

Roles

RoleId	RoleName
1	SuperAdmin
2	ShopAdmin
3	SalesMan

STEP 3: Product & Stock

Products

Field	Type
ProductId	INT (PK)
ShopId	INT
ProductName	NVARCHAR(200)
Barcode	NVARCHAR(50)
SalePrice	DECIMAL(18,2)
IsActive	BIT

Stock

Field	Type
StockId	INT (PK)
ShopId	INT
ProductId	INT
Quantity	DECIMAL(18,2)
LastUpdated	DATETIME

 Stock আলাদা টেবিল রাখলে কন্ট্রোল সহজ হয়

STEP 4: Supplier & Customer

Suppliers

SupplierId	ShopId	SupplierName	Mobile
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Customers

CustomerId	ShopId	CustomerName	Mobile
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STEP 5: Purchase (কেনা)

PurchaseMaster

PurchaseId	ShopId	SupplierId	PurchaseDate	TotalAmount
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PurchaseDetails

PurchaseDetailId	PurchaseId	ProductId	Qty	Price
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 Purchase হলে Stock বাড়বে

STEP 6: Sales (বিক্রি)

SalesMaster

SalesId	ShopId	CustomerId	SalesDate	InvoiceNo	TotalAmount
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SalesDetails

SalesDetailId	SalesId	ProductId	Qty	Price
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 Sales হলে Stock কমবে

STEP 7: Payment (Cash / Bank / bKash)

Payments

PaymentId	ShopId	SalesId	PaymentType	Amount	Date
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PaymentType: Cash / Bank / bKash

STEP 8: Basic Accounting (Simple)

Accounts

AccountId	ShopId	AccountName	Type
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Type: Cash / Bank / Expense / Income

Transactions

TransactionId	ShopId	AccountId	Debit	Credit	Date	RefNo
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STEP 9: SQL Server – CREATE TABLE Script (Multi-Shop Ready)

নিচের স্ক্রিপ্টগুলো **ক্রম অনুযায়ী** রান করবেন। সব টেবিলেই `ShopId` আছে, যাতে মাল্টি-শপ ডাটা আলাদা থাকে।

Shops

```
CREATE TABLE Shops (  
    ShopId INT IDENTITY PRIMARY KEY,  
    ShopName NVARCHAR(200) NOT NULL,  
    OwnerName NVARCHAR(150),  
    Mobile NVARCHAR(20),  
    Email NVARCHAR(100),  
    IsActive BIT DEFAULT 1,  
    CreatedDate DATETIME DEFAULT GETDATE()  
);
```

Roles

```
CREATE TABLE Roles (  
    RoleId INT IDENTITY PRIMARY KEY,  
    RoleName NVARCHAR(50) NOT NULL  
);
```

Users

```
CREATE TABLE Users (  
    UserId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    UserName NVARCHAR(100) NOT NULL,  
    PasswordHash NVARCHAR(MAX) NOT NULL,  
    RoleId INT NOT NULL,  
    IsActive BIT DEFAULT 1,  
  
    CONSTRAINT FK_Users_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),  
    CONSTRAINT FK_Users_Roles FOREIGN KEY (RoleId) REFERENCES Roles(RoleId)  
);
```

Products

```
CREATE TABLE Products (  
    ProductId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    ProductName NVARCHAR(200) NOT NULL,  
    Barcode NVARCHAR(50),  
    SalePrice DECIMAL(18,2) NOT NULL,  
    IsActive BIT DEFAULT 1,  
  
    CONSTRAINT FK_Products_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId)  
);
```

Stock

```
CREATE TABLE Stock (  
    StockId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    ProductId INT NOT NULL,  
    Quantity DECIMAL(18,2) DEFAULT 0,  
    LastUpdated DATETIME DEFAULT GETDATE(),  
  
    CONSTRAINT FK_Stock_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),  
    CONSTRAINT FK_Stock_Products FOREIGN KEY (ProductId) REFERENCES  
Products(ProductId)  
);
```

Suppliers

```
CREATE TABLE Suppliers (  
    SupplierId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    SupplierName NVARCHAR(150) NOT NULL,  
    Mobile NVARCHAR(20),  
  
    CONSTRAINT FK_Suppliers_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId)  
);
```

Customers

```
CREATE TABLE Customers (  
    CustomerId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    CustomerName NVARCHAR(150),  
    Mobile NVARCHAR(20),  
  
    CONSTRAINT FK_Customers_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId)  
);
```

PurchaseMaster

```
CREATE TABLE PurchaseMaster (  
    PurchaseId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    SupplierId INT NOT NULL,  
    PurchaseDate DATETIME DEFAULT GETDATE(),  
    TotalAmount DECIMAL(18,2),  
  
    CONSTRAINT FK_Purchase_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),  
    CONSTRAINT FK_Purchase_Suppliers FOREIGN KEY (SupplierId) REFERENCES  
Suppliers(SupplierId)  
);
```

PurchaseDetails

```
CREATE TABLE PurchaseDetails (  
    PurchaseDetailId INT IDENTITY PRIMARY KEY,  
    PurchaseId INT NOT NULL,  
    ProductId INT NOT NULL,  
    Qty DECIMAL(18,2),  
    Price DECIMAL(18,2),  
  
    CONSTRAINT FK_PD_PM FOREIGN KEY (PurchaseId) REFERENCES  
PurchaseMaster(PurchaseId),  
    CONSTRAINT FK_PD_Products FOREIGN KEY (ProductId) REFERENCES  
Products(ProductId)  
);
```

SalesMaster

```
CREATE TABLE SalesMaster (  
    SalesId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    CustomerId INT NULL,  
    SalesDate DATETIME DEFAULT GETDATE(),  
    InvoiceNo NVARCHAR(50),  
    TotalAmount DECIMAL(18,2),  
  
    CONSTRAINT FK_Sales_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),
```

```
        CONSTRAINT FK_Sales_Customers FOREIGN KEY (CustomerId) REFERENCES  
Customers(CustomerId)  
);
```

SalesDetails

```
CREATE TABLE SalesDetails (  
    SalesDetailId INT IDENTITY PRIMARY KEY,  
    SalesId INT NOT NULL,  
    ProductId INT NOT NULL,  
    Qty DECIMAL(18,2),  
    Price DECIMAL(18,2),  
  
    CONSTRAINT FK_SD_SM FOREIGN KEY (SalesId) REFERENCES SalesMaster(SalesId),  
    CONSTRAINT FK_SD_Products FOREIGN KEY (ProductId) REFERENCES  
Products(ProductId)  
);
```

Payments

```
CREATE TABLE Payments (  
    PaymentId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,  
    SalesId INT NOT NULL,  
    PaymentType NVARCHAR(20), -- Cash / Bank / bKash  
    Amount DECIMAL(18,2),  
    PayDate DATETIME DEFAULT GETDATE(),  
  
    CONSTRAINT FK_Payments_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),  
    CONSTRAINT FK_Payments_Sales FOREIGN KEY (SalesId) REFERENCES  
SalesMaster(SalesId)  
);
```

Accounts

```
CREATE TABLE Accounts (  
    AccountId INT IDENTITY PRIMARY KEY,  
    ShopId INT NOT NULL,
```



```
AccountName NVARCHAR(100),
AccountType NVARCHAR(50), -- Cash / Bank / Expense / Income

CONSTRAINT FK_Accounts_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId)
);
```

Transactions

```
CREATE TABLE Transactions (
    TransactionId INT IDENTITY PRIMARY KEY,
    ShopId INT NOT NULL,
    AccountId INT NOT NULL,
    Debit DECIMAL(18,2) DEFAULT 0,
    Credit DECIMAL(18,2) DEFAULT 0,
    TranDate DATETIME DEFAULT GETDATE(),
    RefNo NVARCHAR(50),

    CONSTRAINT FK_Trans_Shops FOREIGN KEY (ShopId) REFERENCES Shops(ShopId),
    CONSTRAINT FK_Trans_Accounts FOREIGN KEY (AccountId) REFERENCES
Accounts(AccountId)
);
```

Next Step

পরের ধাপে আমরা করবো: ✂️ **ASP.NET Core / Blazor Model Class** ✍️ **Stock Auto Increase / Decrease Logic** 🛏️ **Accounting Posting Logic (Debit / Credit)**

🔔 বলুন, এখন **Option 3 (Model + Business Logic)** এ যাবো?